

A finite-dimensional case of Nowak's kernel-pairing question

FA/Banach automatic discovery run

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Status. Candidate partial result, likely valid. This answers the finite-dimensional case of Remark 4.3 in Nowak's paper [1]; it does not address infinite-dimensional isometric representations.

Source question

The source is Piotr W. Nowak, *Poincare inequalities and rigidity for actions on Banach spaces*, arXiv:1107.1896 [1]. Remark 4.3 asks for which isometric representations the natural map

$$i^* \circ \bar{i} : \ker d_{\bar{\pi}} \longrightarrow (\ker d_{\pi})^*$$

is automatically an isomorphism, or at least surjective. The paper notes that this would eliminate, for such representations, the need for one of the two Poincare-constant inequalities in the proof of the main theorem.

the same time implies that κ_p is finite, which proves the theorem of Proposition 7.1. —

Remark 4.3. Note that under the assumptions of Theorem 1.1, $(\ker d_{\pi})^*$ and $\ker d_{\bar{\pi}}$ are isomorphic (a similar fact holds for $\ker d_{\pi}$ and $(\ker d_{\bar{\pi}})^*$). These spaces are closely related to the spaces of cocycles for the given representations.

It is an interesting question, for which isometric representations is $i^* \circ \bar{i}$ automatically an isomorphism, or at least is surjective. This property would eliminate, for such repre-

sentations, the need to use the inequality $2^{-\frac{1}{p}}\kappa_p < 1$, which is necessary in the above proof. A special case is discussed and applied in Section 7.2.

Remark 4.4. Note that it is not clear whether the above method can be extended to subspaces of $Y \subseteq X$. This would require estimating $\kappa_p(S, X)$ for some p , together with $\kappa_{p^*}(S, X^*)$, where Y^* is a quotient of X^* .

Idea of the proof

Finite-dimensional isometric representations have compact image. Averaging an auxiliary Euclidean or Hermitian inner product over the compact closure of the image makes the representation unitary for that auxiliary inner product. The Riesz map associated with this invariant inner product

intertwines the original representation with the contragredient representation. Applied coordinate-wise to Nowak's cochain spaces, it carries $\ker d_\pi$ onto $\ker d_{\bar{\pi}}$. The source pairing then becomes a positive definite Hilbert sum on the graph of this identification, so no nonzero kernel vector can be annihilated by all dual kernel vectors. Finite dimensionality converts this separation into surjectivity, indeed into an isomorphism.

Definitions

We use the notation of [1]. Let G be generated by a finite symmetric set S equipped with an admissible weight ω on the link graph. Let X be a finite-dimensional normed space and let $\pi : G \rightarrow \text{Isom}(X)$ be an isometric representation. The contragredient representation on X^* is

$$\bar{\pi}_g = \pi_{g^{-1}}^*.$$

The cochain space $C_-^{(1,p)}(G, \pi)$ consists of functions $f : S \rightarrow X$ with $f = -Q_\pi f$, and $d_\pi : C_-^{(1,p)}(G, \pi) \rightarrow C_-^{(2,p)}(G, \pi)$ is Nowak's operator

$$d_\pi f(s, t) = \pi_s f(s^{-1}t) - f(t) + f(s).$$

The map in Remark 4.3 is the restriction-pairing map

$$R_\pi : \ker d_{\bar{\pi}} \longrightarrow (\ker d_\pi)^*, \quad R_\pi(\varphi)(f) = \sum_{s \in S} \varphi(s)(f(s)) \deg_\omega(s).$$

Theorem 1. *Let X be a finite-dimensional real or complex normed space, and let $\pi : G \rightarrow \text{Isom}(X)$ be an isometric representation. Then*

$$R_\pi : \ker d_{\bar{\pi}} \longrightarrow (\ker d_\pi)^*$$

is an isomorphism. In particular, the map $i^ \circ \bar{i}$ in Nowak's Remark 4.3 is automatically surjective for every finite-dimensional isometric representation.*

Proof. The group $\text{Isom}(X)$ is compact in finite dimension: it is a closed and bounded subgroup of $GL(X)$. Hence $K = \overline{\pi(G)}$ is compact. Choose any positive definite Euclidean inner product in the real case, or Hermitian inner product in the complex case, and average it over K . This gives an auxiliary inner product h for which every π_g is unitary.

Define the Riesz identification $J : X \rightarrow X^*$ by

$$(Jx)(y) = h(y, x).$$

In the real case J is linear; in the complex case it is conjugate-linear. In both cases it is a bijection and satisfies

$$J\pi_g = \bar{\pi}_g J, \quad g \in G.$$

Indeed, by K -invariance of h ,

$$(J\pi_g x)(y) = h(y, \pi_g x) = h(\pi_{g^{-1}} y, x) = (\pi_{g^{-1}}^* Jx)(y) = (\bar{\pi}_g Jx)(y).$$

Apply J coordinatewise to functions on S , writing

$$(\mathcal{J}f)(s) = J(f(s)).$$

The intertwining relation shows that $\mathcal{J}$ carries $C_-^{(1,p)}(G, \pi)$ onto $C_-^{(1,p^*)}(G, \bar{\pi})$. It also commutes with the d -operators in the sense that

$$d_{\bar{\pi}}(\mathcal{J}f) = \mathcal{J}(d_{\pi}f),$$

because each term in $d_{\pi}f(s, t) = \pi_s f(s^{-1}t) - f(t) + f(s)$ is sent to the corresponding term in

$$d_{\bar{\pi}}(\mathcal{J}f)(s, t) = \bar{\pi}_s \mathcal{J}f(s^{-1}t) - \mathcal{J}f(t) + \mathcal{J}f(s).$$

Therefore $\mathcal{J}$ is a bijection from $\ker d_{\pi}$ onto $\ker d_{\bar{\pi}}$. In particular, the two kernels have the same finite dimension.

Now let $0 \neq f \in \ker d_{\pi}$. Then $\mathcal{J}f \in \ker d_{\bar{\pi}}$, and the restriction pairing gives

$$R_{\pi}(\mathcal{J}f)(f) = \sum_{s \in S} (\mathcal{J}f(s))(f(s)) \deg_{\omega}(s) = \sum_{s \in S} h(f(s), f(s)) \deg_{\omega}(s) > 0.$$

Thus the range of R_{π} separates points of $\ker d_{\pi}$. Since $\dim \ker d_{\bar{\pi}} = \dim \ker d_{\pi} = \dim(\ker d_{\pi})^*$, this separation implies that R_{π} is surjective, hence an isomorphism. $\square$

Verification notes

The proof is purely finite-dimensional and does not use either Poincare constant inequality from Nowak's main theorem. The compactness input is only the standard fact that the linear isometry group of a finite-dimensional normed space is compact. The complex case uses the conjugate-linear Riesz map only to identify the two kernels and separate points; this is enough for complex-linear surjectivity of R_{π} .

The script `code/finite_dimensional_kernel_pairing_check.py` performs rank sanity checks for cyclic and free-group-style generating sets with non-orthogonal conjugates of rotations. It checks that the restriction pairing has full rank in those examples. These computations are not used in the proof.

Novelty and limitations

Cheap run-index searches for arXiv:1107.1896 and phrases around the finite-dimensional kernel-pairing statement found no duplicate packet. Web searches on 2026-07-03 for exact phrases involving $i^* \circ \bar{i}$, finite-dimensional isometric representations, and the paper title found no prior answer beyond the source paper. The result remains a partial answer: Remark 4.3 is still open here for infinite-dimensional isometric representations.

References

- [1] P. W. Nowak, *Poincare inequalities and rigidity for actions on Banach spaces*, arXiv:1107.1896, 2011.